

BASIC NETWORKING CONCEPTS

Networking Concepts

Network

A group of computer systems connected to share information. The Internet is the largest possible example of this. In a network, all computers can request and send information, but in practice most computers request it and only a few send it out.

For computers to share information over a network, several key components must be in place. Each computer must have network media, a network interface, and network protocols.

Key Components:

- **Network Media:** The pathways that connect computers. In networking, media describes the different ways information is stored and transferred. Computers transmit information over wired connections using coaxial, Ethernet, or fiber-optic cables. Information can also move wirelessly, through radio signals in Wi-Fi. Computers must have either a wired or a wireless connection to be part of a network.
- **Network Interface:** An interface is the point at which two (2) different systems interact with each other. A network interface is the point where a computer connects to the larger network. Each computer needs a network interface, a special port for connecting to the network (via cables) or wirelessly. This is typically done using a piece of hardware called a **network interface card (NIC)**. NICs are often included in the original computer build, but can be added later if the computer does not come with one. Upgrading a system with a NIC or an expansion card after purchase is easy on the motherboard.
- **Network Protocols:** A shared set of protocols, or rules, serves as a common language among computers. These rules are set up at the operating system (OS) level. All the computers on the network must use the same one.

Using Networking Components

A **wired system**, one that uses cables to connect to the network, will most often use Ethernet cabling as its network media. Those cables use specialized sockets and plugs called RJ-45 (Registered Jack Type 45) to make this connection.

The RJ-45 standard makes it easy for industry professionals and for computer manufacturers to create network connections. The RJ-45 plug or clip attaches to the cable ends. One end plugs into the computer, and the other end plugs into the socket on the wall plate.

On the other side of the wall plate is another network cable. At home, this belongs to the internet service provider (ISP). At work, this network cable belongs to the organization and runs to other network devices before connecting to the Internet.

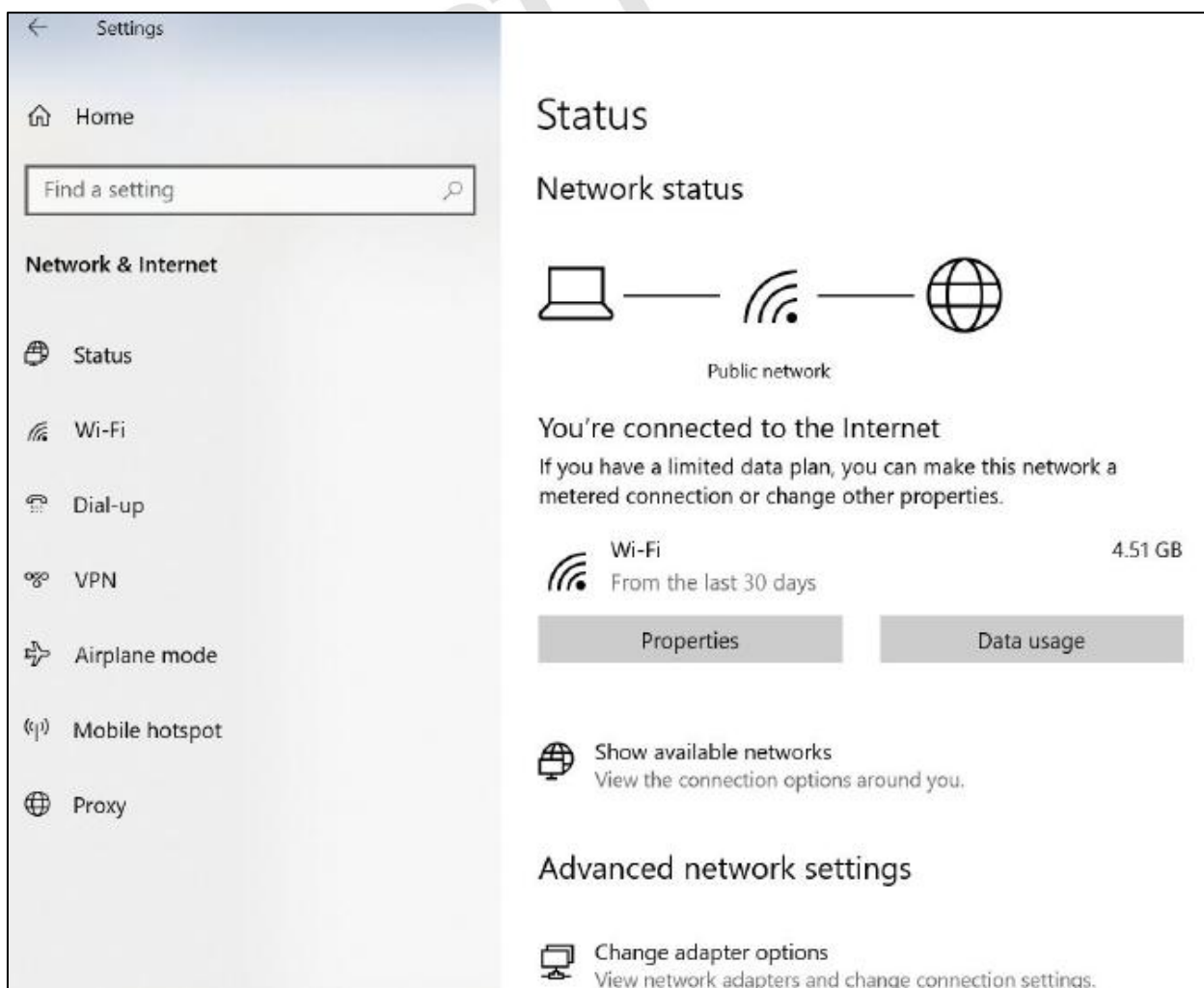
Wireless networks use radio signals instead of cables to communicate. Connecting to a Wi-Fi network is a bit more complex. A computer needs a wireless network interface. Different computers have different ways of connecting to Wi-Fi. Many desktop computers already have built-in Wi-Fi, but others need a **wireless network adapter**. The easiest way to add Wi-Fi to these computers is to plug a small device called a USB wireless adapter into a USB port.

Connecting to a Network

For Windows computers, Microsoft has set up a few ways to check whether the computer is connected to a network or the Internet. One way is to look for a special icon that appears in the bottom-right corner of the screen, in the area called the *system tray*. If the computer is connected with a cable, an icon that looks like a computer will appear. If Wi-Fi is enabled, an icon with curved lines resembling waves appears. The number of waves shows how strong the signal is. If you cannot connect to the internet, the icon changes to a globe with a "no" symbol over it.



Going further into Windows Settings gives more details about the computer's connections. The network settings indicate whether the device is connected via cable or wirelessly, and they also show other options, such as using VPNs, sharing the internet with other devices as a mobile hotspot, or using proxies.



Network Ports and Addresses

Network Ports

When a computer is connected to a network, it can transfer large amounts of information very quickly. The network needs rules, or protocols, to guide traffic to the right place. Network traffic is easier to manage with network identifiers such as IP addresses and ports.

Network Ports are unique numbers that the operating system assigns to different types of network traffic. **Port numbers** help to direct the information on a network, depending on what kind of information it is. For example, when sending an email to a co-worker, the data bundle includes the port number 587. It is like a special ticket that sends the data to a corresponding port on the co-worker's computer.

Some port numbers are reserved for the most common services, such as email and file transfer. Other services get a port number only when in use. Once that service is done using the port, the number is unassigned again. Some types of ports are turned off; closed ports cannot be accessed.

Port Forwarding

Port forwarding can be likened to telling a letter carrier to send the mail to a new address if the receiver is not at home. The receiver still wants to receive the information, even if at a different location. That is essentially what happens in port forwarding: telling the router or firewall what to do with the packets of information.

A common port for this is port 3389, used for the Remote **Desktop Protocol (RDP)** on a graphical user interface (GUI). This basically lets the connection to any other Windows computer anywhere in the world.

To make this happen, set things up correctly on both computers and the internet router, which is a bit like making sure both doors are unlocked and open so one can walk through.

Here are some common port numbers, port number names, and their definition.

Port Number	Name	Definition
21	File Transfer Protocol (FTP)	Standard protocol used for moving files between computers
22	Secure Shell (SSH)	Provides for secure communication between devices
23	Telnet	Provides for unsecure communication between devices
25	Simple Mail Transfer Protocol (SMTP)	Used for sending email. However, it has been deprecated in the last decade
53	Domain Name System (DNS)	Provides a translation between friendly domain names and IP addresses
80	Hypertext Transfer Protocol (HTTP)	Provides communication standards for clients and servers on the World Wide Web
110	Post Office Protocol version 3 (POP3)	Used by email programs for incoming email
143	Internet Message Access Protocol (IMAP)	Used most commonly for receiving email
443	Hypertext Transfer Protocol Secure (HTTPS)	Establishes encrypted communication between clients and servers on the World Wide Web
587	Simple Mail Transfer Protocol (SMTP)	Used for outgoing email

Network Address

Think of a small network, like one in a small office. It is small enough to have only a couple of desktop computers, a printer, and a server. Each of these devices has its own set of addresses that allow it to communicate with the others.

Network addresses are essential components in computer networking, enabling devices to communicate with each other. There are two (2) primary types of network addresses: **physical** and **logical**. Understanding these differences is crucial for network design, troubleshooting, and ensuring efficient communication within and between networks.

Physical Addresses

Every device, whether a computer, a Wi-Fi adapter, or something else, has a unique code called a **Media Access Control (MAC)** address, also known as a *physical address*. This code is unique to each device, hardcoded into the device's network interface card (NIC), and cannot be changed. Here's an example of a MAC address:

70-D9-5B-36-C2-EE

There are six parts, each with two (2) numbers or letters. These two (2) characters are hexadecimal numbers, separated by dashes (-) or colons (:).

The hexadecimal numbers represent a 48-bit binary number. This number is used by switches in a local area network (LAN) to direct packets of data to their proper destination. Each MAC address is unique to each device, hardcoded into the device's network interface card (NIC), and cannot be changed.

Physical addresses have these characteristics:

- **Duration:** Permanent and hardcoded into the NIC.
- **Use case:** Used for local network communication.
- **Notation:** Represented in hexadecimal format.
- **Assignment:** Assigned by the NIC manufacturer.

Logical Addresses

To make sure that every device on a network can be found, each needs its own unique **Internet Protocol (IP)** address. Logical addresses are like special codes that software assigns. Still, these codes can be changed either by a network administrator or by an automated system called the **Dynamic Host Configuration Protocol (DHCP)**.

Logical addresses have these characteristics:

- **Duration:** Can be either static or dynamic and can change over time.
- **Use case:** Used for communication across different networks.
- **Notation:** Represented in either decimal (IPv4) or hexadecimal (IPv6) format.
- **Assignment:** Assigned by network administrators or dynamically by DHCP servers.

One of the most common forms of logical addressing is **IPv4 addresses**. Think of it like a phone number, but for computers and other devices. Just like writing a phone number in a specific way (first the country code or area code, then the exchange prefix, and so on), an IP address has a specific pattern too. It looks like four (4) sets of numbers written with dots in between them, for example:

192.168.1.1

This way of writing is called **dotted decimal notation**. Each part of the address is known as an octet, or group of 8 bits. In IP addressing, a group of 8 bits is an "*octet*," not a byte, because the term "byte" is closely associated with storage. Using "*octet*" makes it clearer when talking about IP addresses.

The IT community will express, or write, an IPv4 address in only one (1) of two (2) ways. In dotted decimal notation, where each octet is a number between 0 and 255:

131.107.2.200

Or in binary. In this case, each octet is an 8-digit binary number, as in this example:

10000011.01101011.00000010.11001000

For most people, dotted decimal notation is easiest to read. This is by far the most common way to write IPv4 network addresses.

Here are some key things to remember about IPv4 addresses:

- They are made of 32 bits
- The address is split into four (4) octets (four (4) groups of 8 bits).
- Each octet is separated by a period, or dot.

IPv6 Addresses

The IPv4 system had been working fine until recently. The problem is that as more devices (such as phones, tablets, and computers) connect to the Internet, the world is running out of IPv4 addresses. Servers, routers, and other Internet-connected devices are moving away from IPv4 addresses to IPv6, because it offers more combinations of unique numbers.

These are a bit different. They are made up of eight (8) sets of four (4) hexadecimal numbers. And that small detail makes a huge difference. While IPv4 could give us over 4 billion different addresses, using IPv6 can give us a huge number of them, up to 1 with 36 zeros after it.

Here is an example of what an IPv6 address looks like:

59C2:339A:0000:4BE5:C12D:6842:1F3E:09E2

It is a long number, 128 bits long, to be exact. This means that by using IPv6 addresses, the global community will not run out of internet addresses anytime soon.

Reference:

CompTIA. (2025). *CompTIA Tech+ CertMaster Learn* [Course module].